LECTURE NOTES

Lecture 12: Optimization in Machine Learning

SDU

Simon Holm
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DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Contents

1 Theoretical Analysis - Preliminaries

We know that for

$$F(w) = \begin{cases} R_n(w) = \frac{1}{n} = \sum_{i=1}^{n} f_i(w) & \text{Emperical Risk} \\ R(w) = \mathbb{E}_{\Xi}[f(w;\Xi)] & \text{Expected Risk} \end{cases}$$

Procedure SG(...)

Choose an inital iterate $\boldsymbol{w_0}$
for $k = 0, 1, ...$ **do**
Generate a realization of the random variable Ξ_k;
Compute a stochastic vector $g(w_k, \Xi; k)$;
Choose a stepsize $\alpha_k > 0$;
Set the new iterate as $w_{k+1} \leftarrow w_k - \alpha g(w_k, \Xi_k)$;

2 Convergence Analysis - Assumptions

$$g(w_k, \xi_k) = \begin{cases} \nabla f((w_k; \xi_k)) \\ \frac{1}{n_k} \sum_{i=1}^{n_k} \nabla f((w_k; \xi_k)) \\ H_k \cdot \frac{1}{n_k} \sum_{i=1}^{n_k} \nabla f((w_k; \xi_k)) \end{cases}$$

3 Convergence Analysis - Results

4 Computational Complexity Analysis